Software Engineering - Lab 4
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Task 1

I am a game developer currently working on a Memory Color Pattern game
The game has:
1) A 2x2 grid of four colored buttons (Red, Blue, Green, Yellow) with dim and illuminated lighting states
2) Automated timed playback that flashes the sequence step-by-step for the player to watch
3) Click detection registering player inputs and checking order against the target sequence
4) Score tracking, visual turn indicators, and a Game Over overlay with restart functionality
The game has a bug and three features left as tasks to implement.
I need you to analyze, and complete/fix the game to make it fully functional and more interesting.
Task 1: Fix the sequence duplication bug
Each round is supposed to append only one new random color to the existing sequence. In game_engine.start_next_round(), the sequence update line performs self.sequence += self.sequence + [new_color] instead of appending new_color to self.sequence. This causes the pattern length to grow exponentially and duplicates previous entries. Correct the sequence addition logic so only one single color index is added each round.
I am attaching all the code files. Finish Task 1 and suggest all the changes required.

Task 2

Task 2: Implement dynamic playback acceleration
Currently, the sequence flashes at a static 450ms flash duration and 200ms pause duration across all rounds. In game_engine.start_next_round(), adjust self.flash_duration and self.pause_duration to decrease gradually as self.score increases (down to a minimum speed threshold like 180ms flash and 80ms pause), making playback faster and more challenging in later rounds.

Task 3

Task 3: Implement sound effects or audio frequencies
The game currently relies entirely on silent visual flashes. Use pygame.mixer or synthesize simple square/sine audio tones mapped to each color ID (e.g., Red: 261Hz, Blue: 329Hz, Green: 392Hz, Yellow: 523Hz). Play the respective tone whenever a button flashes during WATCH mode and when the player clicks it during PLAYER_TURN.

Task 4

Task 4: Implement a round countdown timer
During PLAYER_TURN, the player can currently wait indefinitely before making their move. Add a per-round or per-step countdown timer bar displayed in the HUD. If the timer reaches

0 before the player completes the required sequence, transition the game state to GAME_OVER.

Debugging

Provide a .gitignore file, ensuring that it excludes Python virtual environments, cache directories, and macOS .DS_Store files.
Also, the game window is crashing and closing immediately after the completion of the first round. Please analyze the game loop and state transition logic and provide the corrected code. Add these screenshots too. Make is properly formatted. Do not use multi color text

Screenshots:





